Variational Information Bottleneck

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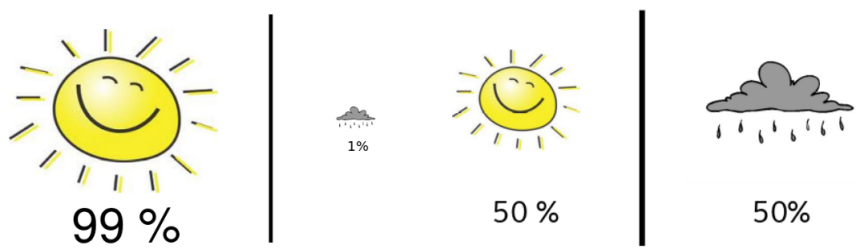
Compression and Learning

Ground breaking advancements in physics can often be summarized in compact equations consisting of only a few terms while being relevant to scales ranging from microscopic to cosmological. For example, Newton's formula for gravity changed the way we think about planetary attraction and insect movement while consisting of 3 terms. Human learning can be thought of as identifying an essential part of a problem and utilizing new found knowledge to find solutions in novel situations. A similar objective can be applied to machine learning. Information theory, a science created by Claude Shannon dealing with subjects such as cryptography and noisy communication formalized essential concepts for quantifying information and uncertainty. Inadvertently, concepts like cross entropy and Kullback–Leibler (KL) divergence are commonplace in today's machine learning workflow. In the presented work *Deep Variational Information Bottleneck* the authors take the subject a step further, and propose information theoretic perspective on machine learning [Alemi et al., 2016]. Specifically they re-imagine the neural network as an information passing problem. The goal is to create an informational bottleneck, maximizing the information between the internal representation of the network and the target while minimizing the mutual information with the input. This is analogous to Newton deriving the gravitational constant devoid of his knowledge of apples. The authors implement a model addressing this satisfying information theoretic perspective while utilizing techniques from variational inference [Kingma and Welling, 2013]. Their final model is an information theoretic reinterpretation of the variational autoencoder demonstrating a novel form of regularization with improved robustness to adversarial attacks.

Information Theory Background

To understand the problem at hand it is first helpful to define key information theory concepts such as entropy, KL divergence, and mutual information. Entropy is used as a measure of uncertainty for a random variable. When Claude Shannon first came up with the formula for entropy he asked John von Neumann what he should name it and received the following reply: "You should call it entropy for two reasons: first because that is what the formula is in statistical mechanics but second and more important, as nobody knows what entropy is, whenever you use the term you will always be at an advantage!"[Batty]. Nevertheless weather prediction systems can provide an intuitive explanation.



Imagine two weather forecasts where one predicts sunshine with 99% probability and rain with 1% while the other predicts sunshine with 50% and rain with 50% probability [Stérin]. Intuitively the first forecast contains less uncertainty. Entropy is quantified with the following formula:

$$H(X) = \sum_{x \in X} p(x) * log(\frac{1}{p(x)})$$

First scenario: $-0.99 * log(0.99) - 0.01 * log(0.01) = 0.0807$

Second scenario: $-0.5 * log(0.5) - 0.5 * log(0.5) = 1$

KL divergence is a measure that is used to calculate a "distance" between two probability distributions. Although it is not a real distance metric as it is non commutative with respect to the inputs,

and so it does not satisfy the triangle inequality, it is still a useful quantification.

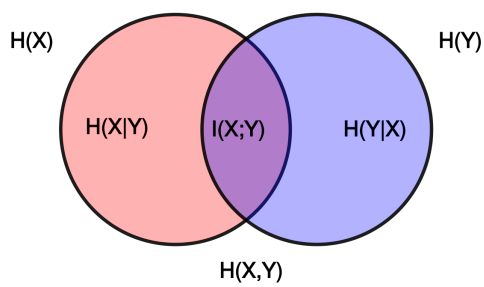
$$D_{KL}(p(X)|q(X)) = \sum_{x \in X} p(x) * log(\frac{p(x)}{q(x)})$$

Important properties of KL divergence include iff p==q then $D_{KL} = 0$ and KL divergence is always non negative.

Finally mutual information denoted as I, is a measure of shared information between two random variables. It can also be written as the KL divergence between the joint probability of two random variables and the product of their individual probabilities:

$$\begin{aligned} I(X;Y) &= \sum_{x \in X} \sum_{y \in Y} p(x,y) * log(\frac{p(x,y)}{p(x) * p(y)} \\ &= D_{KL}(p(X,Y)|p(X)p(Y)) \\ &= H(X) - H(X|Y) = H(X) + H(Y) - H(X;Y). \end{aligned}$$

Another interpretation of mutual information is the sum of individual random variable entropies minus their joint entropy



Variational Information Bottleneck

Information bottleneck was first proposed in 2000 by Tishby et al. The objective of the bottleneck is to identify a representation of the target while being as compressive as possible of the original input. Since the input X, representation Z, and target Y form a Markov chain X $\rightarrow$ Z $\rightarrow$ Y, we can use the data processing inequality to

state p(y| x, z) = p(y | z). This implies that no information is added in the transformation X to Z. Their joint probability can be written as follows

$$p(X, Y, Z) = p(Z|X, Y)P(Y|X)p(X) = p(Z|X)p(Y|X)p(X). \quad (1)$$

Minimizing mutual information between the representation Z and the input X is an important part of the formulation because maximizing information between representation and target can be achieved simply by an identity function from X to Z. This would achieve the first part of the information bottleneck objective, but would not allow for generalizable compressed knowledge representation.

In general, mutual information between encoding Z and target Y has the following formulation:

$$I(Z, Y) = \int p(y, z)\log\frac{p(y, z)}{p(y)p(z)}dydz = \int p(y, z)\log\frac{p(y|z)}{p(y)}dydz \quad (2)$$

which is intractable to compute analytically due to the integration over y. The authors of *deep variational information bottleneck* Alemi 2017 et al apply the information bottleneck objective to neural networks, and so re-frame the learning objective to maximizing the mutual information between an intermediate layer in the network, Z, and target Y while minimizing the mutual information between Z and input X:

$$R_{IB}(\theta) = I(Z, Y; \theta) - \beta I(Z, X; \theta). \quad (3)$$

β is a hyperparameter which scales the regularizing mutual information between Z and X, the part of the equation which encourages the model to forget the input and compress the representation. As mentioned, computing the mutual information is intractable thus the authors instead use variational inference to approximate p(y|z) with q(y|z). Success of this approach was initially demonstrated in the Auto-Encoding Variational Bayes paper [Kingma and Welling, 2013]. Using variational inference we can then rewrite the first term

in our information bottleneck objective as follows

$$I(Z,Y) \geq \int p(y,z)\log\frac{q(y|z)}{p(y)}dydz \tag{4}$$

$$= \int p(y,z)\log q(y|z)dydz - \int p(y)\log p(y)dy \tag{5}$$

$$= \int p(y,z)\log q(y|z)dydz + H(Y). \tag{6}$$

Since H(Y) is constant with respect to the terms we are optimizing we can omit it from the final loss we aim to optimize. Furthermore the term can be rewritten using data processing inequality from (2):

$$I(Z,Y) \geq \int p(x)p(y|x)p(z|x)\log q(y|z)dydzdx. \tag{7}$$

We can also approximate the data and target distribution, p(x, y) using our empirical data resulting in the optimization loss for the first term:

$$I(Z,Y) \geq \frac{1}{N}\sum_{n=1}^{N}\int p(z|x_n)\log q(y_n|z)dz \tag{8}$$

The second, compression term from the bottleneck objective can be refactored in a similar manner.

$$I(Z,X) = \int p(x,z)\log\frac{p(z|x)}{p(z)}dzdx = \int p(x,z)\log p(z|x)dzdx - \int p(z)\log p(z)dz \tag{9}$$

The authors propose a variational approximation, r(x) for the marginal distribution Z and so combining the two terms together results in the following loss

$$L \approx \frac{1}{N}[\sum_{n=1}^{N}\int p(z|x_n)\log q(y_n|z)dz - \beta p(z|x_n)\log\frac{p(z|x_n)}{r(z)}]. \tag{10}$$

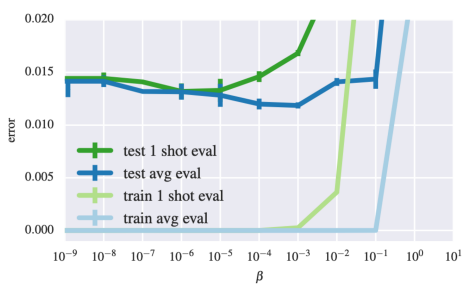
This takes on a familiar form where $p(z|x_n)$ is the posterior i.e. the encoder and $q(y_n|z)$ is the likelihood i.e. the decoder from Kingma Welling 2013. Using the reparameterization trick to avoid back propagating gradients through a stochastic sampler we can rewrite the above loss with a sum over the expectation of ϵ sampled from a

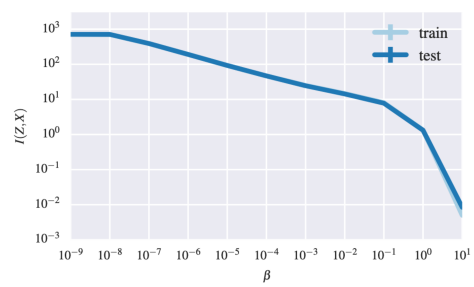
standard Gaussian.

$$J_{IB} = \frac{1}{N}\sum_{n=1}^{N}\mathbb{E}_{\epsilon\ p(\epsilon)}[-\log q(y_n|f(x_n,\epsilon))] + KL[p(Z|x_n), r(Z)]. \quad (11)$$

Empirical Results

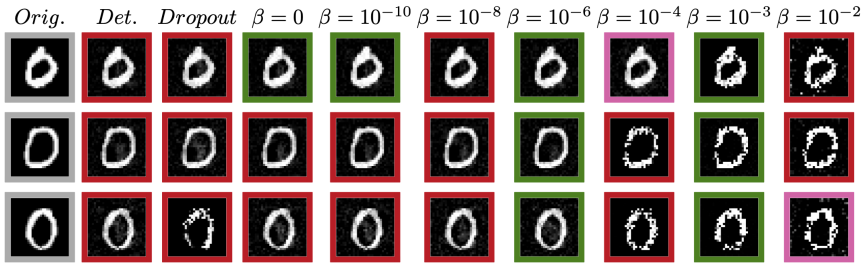
Using the above formulation the authors demonstrate that a simple model consisting of a sigmoid decoder, K-dimensional spherical Gaussian $r(z) = \mathcal{N}(z|0, I)$ and an MLP encoder of the form 784-1024-1024-2K can demonstrate reasonable effectiveness on MNIST achieving a 1.13% accuracy error. An interesting property of this model formulation is its ability to measure mutual information between the encoding, input and, target. On the MNIST example, the authors observe a sharp drop in performance corresponding to $\beta = 10^{-2}$ and mutual information between input and embedding of 10 bits.





Using this fact, we can reliably quantify the complexity of the problem in bits allowing for another way to quantify the intrinsic dimension of the objective [Li et al., 2018].

In addition the authors demonstrate that variational information bottleneck is robust to adversarial attack. They attempted to convert the first 10 zeros from MNIST to 1s using fast gradient sign attack (FGS) [Goodfellow et al., 2014]. Samples that have a green background were not misclassified by the model after an FGS attack.



Conclusions

Variational information bottleneck provides an appealing information theoretic approach to quantify how much a network is learning and how much of the original distribution it is forgetting. It allows the practitioner to quantify the dimensionality of the problem at hand and construct a model more resistant to adversarial attacks. Although far from the state of the art, this work takes an exciting step towards utilizing information theoretic metrics in machine learning. There are other works attempting to quantify what the model is learning during different stages of the process. Another exciting direction is proposed by a recent work from Tishby [Shwartz-Ziv and Tishby, 2017] suggesting that networks have two different training modes: one learning of the target distribution and the other forgetting the input. Overall using mutual information is a natural way to quantify subject matter learned by the model and can be used in the future for deeper model understanding.

References

A. A. Alemi, I. Fischer, J. V. Dillon, and K. Murphy. Deep Variational Information Bottleneck. *arXiv e-prints*, art. arXiv:1612.00410, Dec. 2016.

M. Batty. What von Neumann Said to Shannon. `http://www.spatialcomplexity.info/what-von-neumann-said-to-shannon/`. [Online; accessed 2021].

I. J. Goodfellow, J. Shlens, and C. Szegedy. Explaining and Harnessing Adversarial Examples. *arXiv e-prints*, art. arXiv:1412.6572, Dec. 2014.

D. P. Kingma and M. Welling. Auto-Encoding Variational Bayes. *arXiv e-prints*, art. arXiv:1312.6114, Dec. 2013.

C. Li, H. Farkhoor, R. Liu, and J. Yosinski. Measuring the Intrinsic Dimension of Objective Landscapes. *arXiv e-prints*, art. arXiv:1804.08838, Apr. 2018.

R. Shwartz-Ziv and N. Tishby. Opening the Black Box of Deep Neural Networks via Information. *arXiv e-prints*, art. arXiv:1703.00810, Mar. 2017.

T. Stérin. About Shannon's Entropy. `https://tcosmo.github.io/2019/04/21/shannon-entropy.html`. [Online; accessed 2021].